

On the Convergence of Iterative Methods for Nonlinear Operator Equations

First A. Author*

Second B. Author[†]

May 15, 2026

Abstract

We study the convergence properties of a class of iterative methods applied to nonlinear operator equations in Banach spaces. Under suitable Lipschitz continuity assumptions on the Fréchet derivative, we establish both local and semi-local convergence results. Our main theorem generalises classical results of Kantorovich and provides sharper error bounds.

Keywords: nonlinear equations, iterative methods, Banach space, convergence analysis.

MSC 2020: 65J15, 47H10, 47J25.

1 Introduction

Let X and Y be Banach spaces and let $F: D \subseteq X \rightarrow Y$ be a nonlinear operator. We consider the problem of finding $x^* \in D$ such that

$$F(x^*) = 0. \tag{1}$$

Newton's method generates a sequence $\{x_n\}$ via

$$x_{n+1} = x_n - F'(x_n)^{-1}F(x_n), \quad n = 0, 1, 2, \dots \tag{2}$$

where $F'(x)$ denotes the Fréchet derivative of F at x .

Greek letters appear naturally: the Lipschitz constant γ , step size α , tolerance $\varepsilon > 0$, and domain radius δ .

*Department of Mathematics, University X. Email: `first@university-x.edu`

[†]Institute of Applied Mathematics, University Y. Email: `second@university-y.edu`

2 Preliminaries

Definition 2.1 (Lipschitz Condition). We say $F': D \rightarrow \mathcal{L}(X, Y)$ satisfies a *centre Lipschitz condition* at x_0 with constant $\omega > 0$ if

$$\|F'(x) - F'(x_0)\| \leq \omega \|x - x_0\|, \quad \forall x \in D. \quad (3)$$

Definition 2.2 (ω -Condition). Operator F satisfies the ω -condition on D if there exist constants $\beta, \eta, \omega > 0$ such that

$$\|F'(x_0)^{-1}\| \leq \beta, \quad (4)$$

$$\|F'(x_0)^{-1}F(x_0)\| \leq \eta, \quad (5)$$

$$\|F'(x) - F'(y)\| \leq \omega \|x - y\|, \quad \forall x, y \in D. \quad (6)$$

3 Main Results

Lemma 3.1. Let F satisfy the ω -condition with parameters (β, η, ω) and set $h = \beta\omega\eta$. If $h \leq \frac{1}{2}$, then for every $n \geq 0$,

$$\|x_{n+1} - x_n\| \leq \frac{1}{2^n} \|x_1 - x_0\|. \quad (7)$$

Proof. We proceed by induction on n . The base case $n = 0$ is immediate from the definition of η . Assume the bound holds for some $n \geq 0$. By the Newton step (2) and estimate (3), we obtain

$$\begin{aligned} \|x_{n+2} - x_{n+1}\| &= \|F'(x_{n+1})^{-1}F(x_{n+1})\| \\ &\leq \frac{\omega}{2} \|x_{n+1} - x_n\|^2 \cdot \|F'(x_{n+1})^{-1}\| \\ &\leq \frac{1}{2} \|x_{n+1} - x_n\|, \end{aligned} \quad (8)$$

which completes the induction. $\square$

Theorem 3.2 (Semi-local Convergence). Under the hypotheses of Lemma 3.1, the sequence $\{x_n\}$ defined by (2) converges to a solution x^* of $F(x^*) = 0$. Moreover, the error satisfies

$$\|x_n - x^*\| \leq \frac{2\eta}{2^n - 1}, \quad n \geq 1. \quad (9)$$

Proof. Since $\{x_n\}$ is Cauchy by Lemma 3.1 and X is complete, the sequence converges to some $x^* \in \overline{B}(x_0, r)$ where

$$r = \frac{1 - \sqrt{1 - 2h}}{\omega\beta}.$$

Continuity of F then gives $F(x^*) = 0$. The error bound (9) follows by summing the geometric series from Lemma 3.1. $\square$

Corollary 3.3. If F' is Lipschitz continuous on D and $h < \frac{1}{2}$, then Newton's method converges quadratically:

$$\|x_{n+1} - x^*\| \leq C \|x_n - x^*\|^2, \quad (10)$$

where $C = \frac{\omega\beta}{2}$.

3.1 Matrix Notation Example

The Jacobian of a system $\mathbf{F}: \mathbb{R}^n \rightarrow \mathbb{R}^n$ can be written as

$$J(\mathbf{x}) = \begin{pmatrix} \frac{\partial F_1}{\partial x_1} & \cdots & \frac{\partial F_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial F_n}{\partial x_1} & \cdots & \frac{\partial F_n}{\partial x_n} \end{pmatrix}. \quad (11)$$

3.2 Additional Notation

Fractions and subscripts:

$$a_n = \frac{n!}{\Gamma(n + \alpha + 1)}, \quad S_N = \sum_{k=0}^N \frac{(-1)^k}{(2k+1)!} x^{2k+1}.$$

Operators and Greek letters:

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e, \quad \int_0^\pi \sin \theta \, d\theta = 2.$$

4 Conclusion

We have established semi-local convergence of Newton's method under a centre Lipschitz condition, obtaining error bounds that improve upon the classical Kantorovich theory. Future work will explore higher-order methods and relaxed smoothness assumptions.

References

- [1] L. V. Kantorovich, *On Newton's method for functional equations*, Doklady Akad. Nauk SSSR, **59** (1948), 1237–1240.
- [2] J. M. Ortega and W. C. Rheinboldt, *Iterative Solution of Nonlinear Equations in Several Variables*, Academic Press, New York, 1970.
- [3] I. K. Argyros, *Convergence and Applications of Newton-type Iterations*, Springer, New York, 2008.
- [4] P. Deuffhard, *Newton Methods for Nonlinear Problems*, Springer Series in Computational Mathematics, vol. 35, 2011.